Hints & Solutions

1. Extrinsic semiconductors have more conductivity.

2. Semiconductors are electrically neutral.

3. It is depleted of charge carriers and decreases in forward bias.

5. In N-type semiconductor majority charge carries are electrons.

9. The reverse current is more in Ge because of less band gap.

10. In forward bias potential barrier decreases

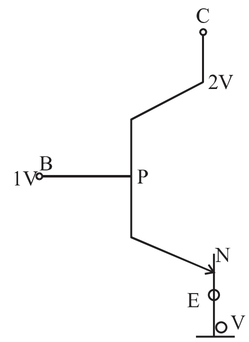
12. electron have higher mobility compared to holes in semiconductors.

13. In sample x no impurity level seen, so it is undoped. In sample y impurity energy level lies below the conduction band so it is doped with fifth group impurity.
In sample z, impurity energy level lies above the valence band so it is doped with third group impurity.

15. Characteristic of p-n junction

16. $f_i = f_0$ for half wave rectifier

17.



$\boxed{V_P - V_N} = 1 - 0 = + \text{ve}$

Base emitter junction = F.B.

$[V_P - V_N] = 1 - 2 = -\text{ve}$

Base collector junction = R.B.

Active region

18. Use the characteristic equations for different gates.

19. $\lambda = \frac{hc}{E} \approx \frac{1240}{1.14} \text{nm} \approx 1088 \text{nm}$